IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 16 August 2024, accepted 14 September 2024, date of publication 23 September 2024, date of current version 16 October 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3466192

RESEARCH ARTICLE

Design of a Personal Credit Risk Prediction Model and Legal Prevention of Financial Risks

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ABSTRACT With the escalating demand for personal credit, banking financial institutions face the imperative to expand their user base and mitigate losses caused by non-performing loans (NPLs). This study addresses the necessity for accurate and efficient personal credit risk assessment systems, focusing on rectifying credit data imbalances and developing a high-performing credit risk prediction model. To tackle data imbalance, deep learning's Generative Adversarial Network (GAN) is employed for oversampling, learning the distribution of default samples within high-risk financial groups. "New" default samples are then integrated into the original dataset to achieve a balanced representation of default and normal samples. In the era of big data, marked by a surge in personal credit data dimensions and sample numbers, an advanced ensemble learning model, the Light Gradient Boosting Machine (LightGBM), is combined with the GAN model for credit risk prediction. The study utilizes six different imbalance ratio datasets from the Knowledge Extraction based on Evolutionary Learning platform to experiment with the GAN model's classification effectiveness. Additionally, consumer finance company credit data, consisting of 559 features, undergoes data preprocessing and feature extraction to validate the LightGBM-GAN model's effectiveness for credit risk prediction. Experimental results reveal that: (1) GAN significantly improves risk information classification (achieving an 86.7% accuracy), outperforming traditional oversampling methods on multiple datasets; (2) The LightGBM-GAN model excels in Area Under the Receiver Operating Characteristic Curve and Kolmogorov-Smirnov statistic values (averaging 0.86 and 0.87, respectively), establishing itself as an effective credit risk prediction model; (3) Feature importance identified by the LightGBM model underscores the predictive potential of financial weakly correlated variables, such as consumer behavior data. This study strives to equip financial institutions with efficient and precise credit risk assessment tools, ensuring compliance with laws and regulations, averting legal risks, reducing NPL risks, and fortifying the stability of the financial system.

INDEX TERMS Risk assessment, risk prediction, legal prevention, generative adversarial network, light gradient boosting machine.

I. INTRODUCTION

In the dynamic landscape of the financial market, the escalating demand for personal credit underscores the imperative to establish a swift and precise personal credit risk assessment system [1], [2]. Financial institutions are tasked not only with fulfilling customers' credit needs but also with mitigating risks and losses stemming from non-performing loans (NPLs) [3]. This challenge extends beyond mere business expansion; it necessitates addressing data imbalances and implementing a credit risk prediction model with robust forecasting capabilities [4], [5]. Data imbalance arises when the number of samples in certain categories is significantly lower than that in the majority, potentially leading to biased model training favoring the majority categories and impeding accurate risk predictions for minority categories [6]. Given the complexity of constructing prediction models that involve numerous features and samples, exploring more effective

The associate editor coordinating the review of this manuscript and approving it for publication was Amjad Mehmood.

methods to surmount these challenges becomes imperative. Zahid and Joudar emphasized that despite advancements in artificial intelligence (AI) capabilities, challenges persisted in information acquisition, significantly impacting performance outcomes. The study underscored that optimizing the information retrieval process could bolster AI systems' adaptability in handling intricate tasks, including credit risk assessment, where data scarcity and imbalance pose substantial challenges. Enhancements in data acquisition and generation technologies were pivotal in augmenting predictive model efficacy [7]. Emerging technologies like Generative Adversarial Networks (GAN) show substantial promise in mitigating data imbalance issues and enhancing model prediction capabilities. GAN operate through adversarial training between a generator and a discriminator, enabling the generation of realistic data samples [8]. This approach is particularly advantageous in enriching datasets with scarce samples from minority classes, thereby improving predictive accuracy in credit risk assessment. GAN have demonstrated effectiveness in fields such as image generation and data augmentation, offering financial institutions more robust tools for credit risk evaluation [9]. However, Albahri et al. suggested that while GAN contributed to innovation and practical problem-solving, they also introduced challenges and risks. Despite their ability to produce high-quality synthetic data for refining model training and prediction, concerns about data privacy and security persist. Financial institutions must carefully balance these considerations when integrating GAN technology to ensure enhanced accuracy and reliability in credit risk assessment [10].

To tackle the challenges outlined above, this study presents solutions substantiated by empirical evidence. Firstly, the GAN in deep learning is introduced as a data oversampling method. This approach effectively rectifies imbalanced datasets by learning the sample distribution of underrepresented categories and generating new default samples. Secondly, a robust credit risk prediction model is devised by integrating the ensemble learning model Light Gradient Boosting Machine (LightGBM) with GAN. This combination adeptly handles high-dimensional features and copious samples within a big data environment. To assess the efficacy of these methodologies, extensive experiments are conducted on both public datasets and authentic consumer finance data. A comprehensive evaluation, involving comparisons of classification effects, prediction performances, and feature importance analyses among various oversampling methods, sheds light on the strengths and limitations of each approach.

This study addresses the pressing need to manage the escalating demand for personal credit while mitigating the risks of NPLs encountered by financial institutions. By tackling the challenges of imbalanced credit data and constructing precise prediction models, the study offers a novel solution. Through the integration of GAN and the LightGBM model, it provides a pioneering method for achieving more accurate and efficient individual credit risk assessment. This approach furnishes reliable support for risk management and decision-making in financial institutions, bolstering operational capabilities, and risk control while aligning with regulatory requirements. Moreover, it aids in reducing the incidence of NPLs, thereby promoting stability and sustainable development within the financial system. Additionally, the methodologies employed in this study yield valuable insights for future research and practice in financial risk management.

The methodological and experimental innovations of this study are demonstrated through several key advancements. First, the introduction of the Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) addresses common challenges such as mode collapse and vanishing gradients found in traditional GAN. By utilizing the Wasserstein distance and incorporating gradient penalty mechanisms, this approach significantly enhances data quality and improves the reliability of model training. Second, the study combines high-quality samples generated by WGAN-GP with LightGBM to develop an efficient credit risk prediction model. This combination shows substantial improvements in prediction accuracy and stability, particularly in financial data analysis. The study further breaks new ground by integrating GAN technology with financial risk prediction, exploring innovative methods for handling complex financial datasets through the fusion of GAN and LightGBM. This approach opens new avenues for future research. Extensive experimentation on both public datasets and real-world consumer financial data confirms the robustness and generalizability of the model across various data environments, underscoring the practical applicability of the proposed framework. Additionally, feature importance analysis identifies several critical factors essential to default prediction, thereby enhancing the interpretability of the model. This provides financial institutions with actionable insights to improve decision-making processes and strengthens overall risk management practices. These innovations offer valuable contributions to the field of credit risk prediction, enriching the analytical toolkit for financial data analysis and demonstrating both theoretical advancements and practical application.

II. LITERATURE REVIEW

In recent years, research in risk prediction has markedly progressed through the integration of machine learning technologies. These studies encompass diverse methodologies, including Support Vector Machine (SVM), Decision Tree, Neural Network, and Ensemble Learning, applied within the financial sector, particularly in credit scoring models. Furthermore, these technologies have been extensively investigated in practical applications such as supply chain financing and credit card fraud detection. They not only overcome challenges encountered by traditional models in terms of dimensionality, model robustness, and accuracy but also catalyze the advancement of risk management and decision-making processes. In the realm of SVM, Li and Fu innovatively combined SVM with T-test feature weighting and fuzzy intra-class distribution to develop a credit scoring model. This approach enhances model robustness by streamlining kernel selection and parameter tuning, addressing dimensionality challenges [11]. Similarly, Shi et al. introduced a Support Vector Regression (SVR) model with kernel function, demonstrating superior accuracy and efficiency compared to logistic regression (LR) and NN counterparts [12]. Notably, Fallah Shams et al. leveraged information entropy for feature selection and employed SVM in predicting supply chain financing enterprises, yielding optimal outcomes in comparative experiments. SVM emerges as a viable option for risk prediction in scenarios with limited sample sizes, particularly within enterprise contexts [13]. In the domain of DT, Yao et al. proposed a novel cost-sensitive DT approach, minimizing misclassification costs while determining node split attributes. Applied in credit card fraud detection, this method outperformed traditional approaches, showcasing its efficacy [14]. Pandey and Bandhu established DT, K-Nearest Neighbors (KNN), and LR models for listed companies, highlighting the superior overall accuracy of DT models based on experimental analysis [15]. Gnoatto et al. devised a two-stage model leveraging NNs. The first stage involved constructing a credit scoring model driven by artificial neural networks, followed by employing case-based reasoning for classification, effectively reducing class I errors. This model demonstrated accurate credit scoring and potential for user recovery guidance [16]. The integration of GAN and LightGBM in credit risk prediction has garnered significant interest. GAN augment model robustness by generating realistic data samples, such as synthesizing loan application data to enhance assessment accuracy [17]. LightGBM, renowned for its efficacy with large-scale datasets and capturing complex feature relationships, is instrumental in feature engineering and model training to boost predictive performance [18]. Mahajan et al. implemented the LightGBM algorithm in credit risk assessment, showcasing enhanced accuracy compared to single decision tree models, along with efficient training and minimal memory consumption [19].

Research on ML algorithms for credit risk prediction can be categorized into two main approaches. Firstly, the introduction of new algorithms like neural networks aims to enhance prediction accuracy. Secondly, the integration of traditional ML algorithms, such as decision tree-based ensemble models, aims for improved interpretability and classification performance. While neural networks excel in accuracy, they lack interpretability, whereas decision tree-based ensemble models offer both classification performance and interpretability. This study seeks to explore the potential of incorporating cutting-edge boosting ensemble algorithms into personal credit risk prediction. Additionally, emerging algorithms like GAN and LightGBM show promise in this field. GAN enhance data realism, while LightGBM captures intricate feature relationships, demonstrating strong performance and interpretability. Therefore, this study reviews the application of GAN and LightGBM in personal credit risk prediction, providing a comprehensive research perspective and inspiration.

III. THEORETICAL BASIS AND MODEL IMPLEMENTING

A. CREDIT IMBALANCE DATA PROCESSING USING GAN

Addressing the imbalance in personal financial credit default samples involves opting for data oversampling. Currently, GAN stands out as a generative model designed to generate data. This study leverages GAN to synthesize tabular data, enhancing the model's classification efficacy.

1) GAN AND ITS VARIANTS

Implementing a GAN model requires careful consideration of the network structure and loss functions for the generator and discriminator. Varied choices in each aspect result in numerous GAN variants [20], [21]. Generally, GAN follow two improvement directions: those tailored to solve specific problems and those enhancing the overall GAN model [22]. The original GAN adopts Jensen-Shannon Divergence as its loss function, presenting a constant metric value when samples lack common elements. This can lead to gradient vanishing issues, making convergence challenging. Therefore, this study opts to construct a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) [23]. The subsequent sections outline the specifics of GAN, Wasserstein Generative Adversarial Network (WGAN), and WGAN-GP.

a: GAN

GAN represents a structural framework rather than a specific network model [24]. The network structure is illustrated in Figure 1.

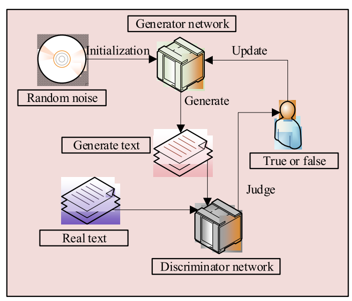

FIGURE 1. Overall structure of GAN.

Figure 1 illustrates that the GAN structure primarily comprises two components: the "generator" and the "discriminator" [25]. The generator is tasked with producing data that conforms to the real sample distribution, unrestricted in form [26]. Meanwhile, the discriminator functions as a dynamically self-updating objective, aiming to minimize the probability of identifying generated samples as true and minimize the likelihood of labeling real samples as false [27].

b: WGAN

WGAN enhances GAN by adopting the Wasserstein distance as the measurement between data distributions instead of traditional metrics [28], [29]. Compared to the original GAN, WGAN introduces the following modifications:

a. The removal of the logarithmic operation from the original GAN's objective function.

b. Elimination of the sigmoid activation function in the discriminator's output layer structure. In the original GAN, the discriminator's task involves classifying real samples from generated samples, constituting a classification task. In contrast, WGAN reformulates the discriminator's task to address the Wasserstein distance between real and generated samples, constituting a regression task.

c. The control of the floating range of parameter 'w' to ensure the discriminator's simulated function adheres to the Lipschitz condition.

WGAN offers significant advantages over traditional GAN and their variants. Firstly, WGAN enhances training stability by employing the Wasserstein distance (also known as Earth-Mover distance) instead of traditional Jensen-Shannon divergence or Kullback-Leibler divergence. The Wasserstein distance provides smoother and more continuous gradients, facilitating stable and meaningful gradient information for both the generator and discriminator during training. This approach mitigates issues such as gradient vanishing or exploding. Secondly, WGAN effectively addresses the problem of mode collapse, which is prevalent in traditional GAN that tend to produce limited sample patterns. WGAN achieves broader coverage of the training data distribution through a more appropriate distance metric. Thirdly, WGAN's loss function is more intuitive, directly reflecting the distance between the generated data distribution and the real data distribution. This characteristic offers clear guidance for optimizing the model. These advantages enable WGAN to excel in generating diverse and high-quality samples, positioning it as a critical tool for addressing data imbalance and enhancing model prediction capabilities.

c: WGAN-GP

In recent years, research on GAN has witnessed significant advancements, resulting in the development of several notable variants and improvements. Style-GAN, for instance, enhances the quality of image generation by introducing "style" controls that allow manipulation of different levels of image features during the generation process. Big-GAN achieves superior resolution and generation outcomes by scaling up model capacity and dataset size. Self-Attention GAN integrates self-attention mechanisms to capture long-range dependencies, thereby producing more consistent and high-quality images. Cycle-GAN introduces an unsupervised image-to-image translation method that facilitates domain adaptation without the need for paired training data, relying on cycle consistency loss. These innovations underscore the extensive potential of GAN in terms of enhancing diversity and performance capabilities in image generation tasks.

This study adopts WGAN-GP as the primary method due to its notable advantages in stability and efficacy within the domain of GAN. WGAN-GP extends the foundation of WGAN by incorporating a gradient penalty, which enhances the stability of the training process significantly. By imposing this penalty, WGAN-GP ensures that discriminator gradients remain within an optimal range, thereby mitigating issues such as gradient explosion or vanishing. These enhancements contribute to a more consistent and dependable training procedure, particularly beneficial when handling complex datasets. Consequently, WGAN-GP excels in generating high-quality and diverse samples, which proves essential for tackling data imbalance challenges and enhancing model prediction capabilities. Hence, WGAN-GP is selected as the primary method for this study.

WGAN-GP improves upon WGAN by introducing a gradient penalty term for each sample in the objective function, facilitating quicker convergence and ensuring compliance with the Lipschitz condition. It replaces gradient clipping in WGAN, resulting in improved training speed and sample quality [30], [31].

The structure of WGAN-GP is displayed in Figure 2.

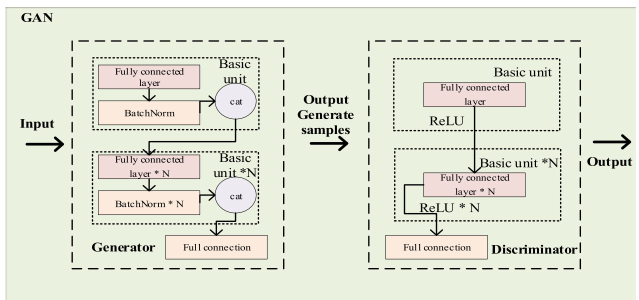

FIGURE 2. **Structural diagram of WGAN-GP.**

Figure 2 illustrates that the generator component of WGAN-GP comprises multiple residual blocks [32]. The input is a noise vector, and in the hidden layer, the fully connected (FC) layer and Batch Norm layer serve as basic units, employing the Rectified Linear Unit (ReLU) activation function. The input and output of each layer are dimensionally spliced for the subsequent layer, implicitly augmenting the feature dimension of the sample. The output layer uses a simple FC layer without activation functions, generating the sample output. In the discriminator, the input dimension matches the output dimension of the generator. The discriminator's hidden layer utilizes full connection, the Leaky ReLU function, and Dropout as basic units, with multiple basic units connected. The output layer employs the FC layer, devoid of any activation functions, and possesses a 1-dimensional output. Additionally, both generators and discriminators leverage Adam as the gradient descent optimizer [33].

2) GAN-BASED OVERSAMPLING METHOD

GAN serves as a generative model, with its pivotal component being the generative network embedded within the two constituent networks. This network has the capability to learn and generate data that adheres to a specific distribution. In the context of credit risk data, the generator is employed to learn and create "new" default samples that align with the distribution of default samples. These generated samples are then added to the original sample set to address the imbalance by augmenting the fewer samples, as depicted in Figure 3.

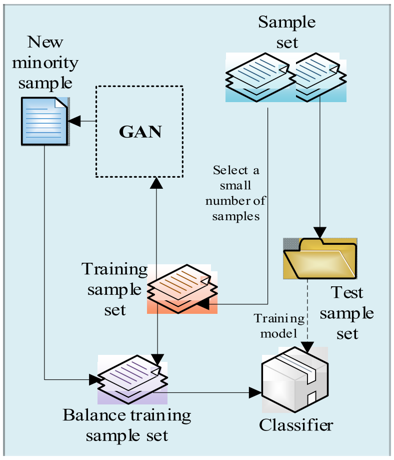

FIGURE 3. Oversampling method based on GAN.

To mitigate the risk of overfitting synthetic data and avoid unintended biases, this study employs several strategic approaches. Firstly, the design of the generator network is carefully managed to regulate its complexity, thereby preventing overfitting to specific patterns within the training data. Secondly, a gradient penalty is integrated into the discriminator's loss function to impose constraints on the gradient range. This measure enhances stability by preventing issues such as gradient explosion or vanishing gradients, enabling the discriminator to effectively distinguish between generated and real samples. Consequently, it facilitates the generator in learning a more accurate data distribution. Lastly, rigorous quality assessment and validation protocols are implemented to ensure that generated samples adhere closely to the statistical characteristics of real data. This rigorous approach not only enhances the performance of risk prediction models with generated data but also safeguards against the introduction of unnecessary biases or spurious patterns.

3) EXPERIMENTAL DESIGN

a: TRAINING OF WGAN-GP

The Variational Autoencoder (VAE) is employed as the generator in the WGAN-GP model to ensure the high quality of generated samples and significantly enhance the model's anomaly detection capabilities. The model architecture is outlined as follows:

· The generator's input is a latent variable, which is derived from the input data via the VAE's encoder and produced using a reparameterization technique. Initially, the encoder in the generator consists of multiple convolutional layers (for image data) or fully connected layers (for other data types) to extract latent features from the data and generate the mean and variance of the latent variables. Subsequently, the latent variables are transformed into normal distributions through reparameterization techniques and passed on to the decoder. The decoder, which comprises either a deconvolution layer (for processing image data) or a fully connected layer (for processing other data types), gradually reconstructs the latent variables back into the original data space. Finally, the decoder outputs the reconstructed data, which maintains the same dimensions as the input data.

· The discriminator's inputs consist of the samples generated by the generator and the true samples. Initially, the input data undergoes extraction through multiple fully connected layers or convolutional layers, each followed by a Leaky ReLU activation function and a batch normalization layer. This configuration enhances the model's expressive power and training stability. The final layers of the network compress the feature map into a one-dimensional output, yielding a probability value via the sigmoid activation function, which indicates whether the input sample is genuine. To further enhance stability, the discriminator incorporates a gradient penalty term that constrains the gradient range, preventing issues like gradient explosion or vanishing. This addition improves both the robustness of the training process and the quality of the generated samples.

· The optimizer employed is the Adam algorithm, known for its adaptive learning rate, which effectively adjusts the parameters of both the generator and the discriminator. The loss functions for the generator and the discriminator are optimized using different strategies to enhance performance. For the discriminator, the Wasserstein loss function, combined with a gradient penalty, is utilized to measure the distance between generated samples and real samples. This approach addresses the common issue of gradient vanishing in traditional GAN, thereby ensuring the model's stability. Specifically, the discriminator's loss function includes the Wasserstein distance between the generated and true samples, while the gradient penalty term ensures the Lipschitz continuity of the discriminator. The generator's loss function, on the other hand, is derived from the discriminator's evaluation of the generated samples. This function encourages the generator to produce more realistic samples by minimizing the Wasserstein distance between the generated and real samples. Through this optimization mechanism, both the generator and discriminator continuously improve their performance during adversarial training.

In the training of GAN, the generator refines its parameters based on the feedback from the discriminator. The discriminator, in turn, optimizes its parameters by evaluating both generated and real samples. The training procedure of WGAN-GP unfolds as follows:

Step 1: For each batch in every epoch, random noise of batch size is drawn from a normal distribution.

Step 2: The generator is employed to generate a batch of synthetic samples.

Step 3: A batch of real samples is selected.

Step 4: The discriminator computes the respective values for real and generated samples, averaging them and calculating the gradient penalty term.

Step 5: The discriminator loss function is optimized via gradient descent, leading to an update in the discriminator parameters.

Step 6: Another batch of random noise is sampled from the normal distribution.

Step 7: The generator generates another batch of synthetic samples.

Step 8: The updated discriminator is utilized to produce the final synthetic sample.

This cyclic process continues iteratively, contributing to the iterative refinement of the WGAN-GP model.

b: EXPERIMENTAL CONTENTS

i) LOSS EVALUATION OF WGAN-GP

The loss of WGAN-GP is meticulously assessed throughout the experiment.

i) CLASSIFICATION ALGORITHM VALIDATION

The adaptability of the data oversampling method based on WGAN-GP to diverse classification algorithms is rigorously examined. This involves the selection of multiple classification algorithms, including traditional LR, SVM, and DT, for thorough testing. Additionally, to scrutinize the impact of the GAN-based oversampling method on classification outcomes, the experiment incorporates alternative oversampling methods for comprehensive testing and comparison. In this regard, Random Over-Sampling (ROS) is specifically chosen as the benchmark oversampling method for comparative analysis [34].

c: DATASET SELECTION

The experimental datasets utilized in this study are sourced from the Knowledge Extraction based on Evolutionary Learning (KEEL) database. The KEEL database stands as an open repository extensively employed in ML and research on imbalanced data processing, encompassing a diverse array of imbalanced datasets. Notably, the accessibility of the KEEL database as a public resource obviates the need for specific permissions or authorizations to obtain and utilize the data for research purposes. Consequently, researchers can freely procure the requisite datasets from the KEEL database for experimentation and scholarly inquiry.

To assess the efficacy of the WGAN-GP-based oversampling method across diverse datasets and various balance ratios, this study curated six datasets sourced from the KEEL database spanning from March 3, 2021, to June 23, 2021. These datasets exhibit varying levels of class imbalance, ranging from 2 to 58. The specific dataset details are outlined in Table 1, encompassing information such as dataset names, imbalance ratios, sample sizes, and feature counts. Notably, the imbalance ratio denotes the proportion of majority class samples to minority class samples.

TABLE 1. Specific information of KEEL imbalanced dataset.

Datasets	Imbalance ratio	Sample size	Number of features
yeast1	2.46	1484	8
ecoli2	5.46	336	7
page-blocks0	8.79	5472	10
abalone9-18	16.4	731	8
winequality-red-4	29.17	1599	11
poker-8-9_vs_6	58.4	1485	10

Subsequently, the aforementioned datasets are partitioned into training and test sets in a ratio of 7:3.

d: EXPERIMENTAL INDICATORS

The assessment of classifier performance employs the classification accuracy (%), a pivotal metric indicating the effectiveness of the classifier across diverse thresholds. The accuracy value serves as a comprehensive measure of the classifier's performance under varying conditions.

B. CREDIT RISK PREDICTION MODEL BASED ON ENSEMBLE LEARNING

1) CREDIT DATA ANALYSIS

The dataset utilized in this study originates from a credit default prediction competition hosted by a specific company and the Kaggle competition platform. The objective of this competition aligns with the research aim of predicting customers' repayment capacity through diverse information, including credit and transaction data, employing statistical or ML methods. Comprising seven data files, the dataset encompasses user application forms, credit information center datasheets, total credit record sheets, and specific behavior record sheets.

2) ESTABLISHMENT OF PERSONAL CREDIT RISK PREDICTION MODEL

a: LightGBM

LightGBM, introduced by Microsoft Research Asia in 2017, is a boosting algorithm that employs histograms instead of feature sorting to reduce the number of feature split points. It utilizes the Gradient-based One-Side Sampling algorithm for data sampling, effectively reducing sample size. Additionally, the Exclusive Feature Bundling algorithm is utilized to bundle features and decrease the number of features [35].

b: MODEL PARAMETER SETTING

During the hyperparameter tuning of the WGAN-GP model, the LightGBM model is first meticulously adjusted. Hyperparameters such as the number of leaves, learning rate, and number of estimators are carefully set to ensure effective learning from the training data while controlling model complexity to prevent overfitting. In the WGAN-GP model, the Adam optimizer and gradient penalty mechanism are employed to enhance training stability. The training process utilizes a loop nesting strategy. After each training iteration of the generator, the discriminator is trained multiple times. This approach involves sampling from both real and generated data to calculate the gradient penalty term, preventing the discriminator's weights from becoming overly concentrated. By stabilizing the training of both the generator and the discriminator, this method improves the quality of the generated samples and enhances the overall detection performance of the model. A LightGBM model is constructed to train the credit data, and the optimal settings of model parameters are outlined in Table 2.

TABLE 2. Meaning of model parameters and their optimal settings.

Name of parameter	Symbol	Optimal value setting
Learning rate	learning_rate	0.1
Maximum depth	max_depth	4
Maximum number of leaf nodes	max_leaves	55
Minimum leaf node weight	min_child_weight	5
Sample sampling score	bagging_fraction	0.9
Feature sampling score	feature_fraction	0.9
L1 regularization	-	10^{-5}
L2 regularization	reg_lambda	100

3) EXPERIMENTAL DESIGN

a: EXPERIMENTAL CONTENTS

During the training process of the LightGBM-GAN model, data preprocessing is initially carried out, including the imputation of missing values and feature selection. Given the highly imbalanced nature of the dataset, with a disproportionately low number of minority class samples, it is essential to augment these samples. The LightGBM-GAN method is employed to generate additional minority class samples, thereby achieving a more balanced dataset. The training phase involves several key steps. First, noise samples are extracted from a normal distribution. These noise samples are then fed into the generator to produce synthetic samples. The discriminator is trained using real samples and evaluates the scores of both real and generated samples. A gradient penalty term is computed to optimize the discriminator's loss function through gradient descent. The generator, guided by the updated discriminator, generates new synthetic samples, and the iterative training process continuously refines the parameters of both the generator and the discriminator. For validation and testing, a 5-fold cross-validation approach is used to train the primary learner. The trained dataset is then input into a logistic regression model for final detection, ensuring the model's accuracy and effectiveness.

a. This study aims to validate the effectiveness of the proposed LightGBM-GAN model for credit risk assessment using the WGAN-GP over-sampling dataset. The performance of the LightGBM-GAN model is compared with advanced models such as the Gradient Boosting Decision Tree (GBDT) algorithm, Stacking model, and Graph Sample and Aggregation (GraphSAGE) model.

b. Additionally, it is necessary to verify whether incorporating WGAN-GP into the study can enhance the model's performance based on the original dataset.

b: EXPERIMENTAL INDICATORS

In this experiment, recall, Area under Curve (AUC), and Kolmogorov-Smirnov (KS) values are utilized as evaluation metrics for the model.

c: EXPERIMENTAL ENVIRONMENT

The experimental setup utilized a computer equipped with an Intel Core i7-10700K CPU and 32GB DDR4 memory. The operating system used is Windows 10 64-bit, with Python 3.8.5 serving as the programming language. Essential libraries, including TensorFlow 2.6.0 and Scikit-learn 0.24.2, arw employed as dependencies.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. DATA OVER-SAMPLING EFFECT OF THE WGAN-GP MODEL

1) LOSS ANALYSIS OF THE WGAN-GP MODEL

Upon experimental verification, the loss of the WGAN-GP model is depicted in Figure 4.

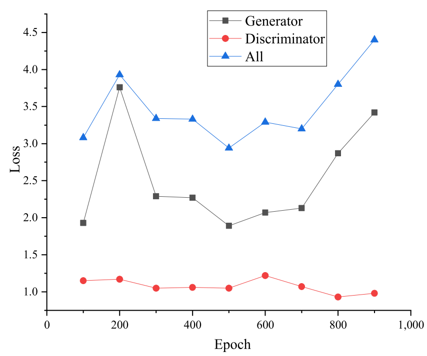

FIGURE 4. Loss result of the WGAN-GP model.

Figure 4 illustrates the training dynamics of the WGAN-GP model, highlighting the changes in loss values for both the generator and the discriminator. As the number of iterations increases, fluctuations in the loss values of both the generator and the discriminator are observed. During the initial 100 iterations, the generator's loss is recorded at 1.93, while the discriminator's loss is 1.15, resulting in a total loss of 3.08. By the 200th iteration, the generator's loss significantly increases to 3.76, with the discriminator's loss slightly rising to 1.17, leading to a total loss of 3.93. In the subsequent iterations, from 300 to 500, the generator's loss decreases to 2.29, 2.27, and 1.89, respectively, while the discriminator's loss remains relatively stable at approximately 1.05. The total loss during this phase slightly decreases to 2.94. However, between 600 and 900 iterations, the generator's loss fluctuates again, reaching a maximum of 3.42 and a minimum of 2.07. The discriminator's loss varies between 0.93 and 1.22, causing the total loss to fluctuate between 3.29 and 4.4. Overall, the discriminator's loss in the WGAN-GP model remains close to zero, indicating that the discriminator is effectively distinguishing between the generated and real samples. The generator's loss remains relatively stable, fluctuating between 1.5 and 3.5, suggesting that the generator is performing well throughout the training process.

2) CLASSIFICATION PERFORMANCE OF VARIOUS MODELS UNDER DIFFERENT OVER-SAMPLING METHODS

Following experiments, the classification outcomes of diverse models under various over-sampling techniques across multiple datasets are presented in Figures 5 and 6.

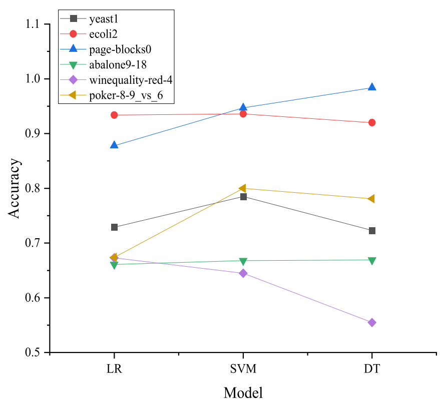

FIGURE 5. Classification results (accuracy) of various classification models in the KEEL dataset under ROS method.

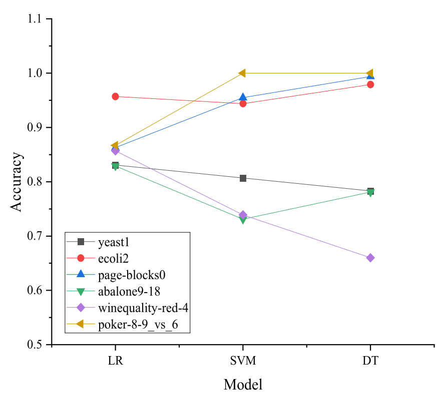

FIGURE 6. Classification results (accuracy) of diverse classification models in the KEEL dataset under WGAN-GP sampling method.

Table 3 lists the classification comparison results of various models under ROS over-sampling and WGAN-GP over-sampling for the KEEL dataset.

TABLE 3. Classification comparison results of different models under ROS and WGAN-GP over-sampling on the KEEL dataset.

Sampling method	Modelling	yeast1	ecoli2	page-blocks0	abalone9-18	winequality-red-4	poker-8-9_vs_6
	LR	0.729	0.934	0.878	0.661	0.673	0.674
ROS	SVM	0.785	0.936	0.947	0.668	0.645	0.8
	DT	0.723	0.92	0.984	0.669	0.555	0.781
	LR	0.831	0.957	0.863	0.829	0.857	0.867
WGAN-GP	SVM	0.807	0.944	0.955	0.731	0.739	1
	DT	0.783	0.979	0.994	0.781	0.66	1

Figure 5 illustrates the performance of various classification models on different datasets under the ROS oversampling mode, highlighting the models' effectiveness in handling imbalanced data. For logistic regression, the classification accuracies across the datasets are as follows: 0.729 for yeast1, 0.934 for ecoli2, 0.878 for page-blocks0, 0.661 for abalone9-18, 0.673 for winequality-red-4, and 0.674 for poker-8 9_vs_6. Logistic regression achieved its highest accuracy of 0.934 on the ecoli2 dataset and its lowest accuracy of 0.661 on the abalone9-18 dataset. Support Vector Machine (SVM) performance is recorded as 0.785 for yeast1, 0.936 for ecoli2, 0.947 for page-blocks0, 0.668 for abalone9-18, 0.645 for winequality-red-4, and 0.800 for poker-8 9_vs_6. SVM exhibited the highest classification accuracy of 0.947 on the page-blocks0 dataset and the lowest accuracy of 0.645 on the winequality-red-4 dataset. For the decision tree model, accuracies are: 0.723 for yeast1, 0.920 for ecoli2, 0.984 for page-blocks0, 0.669 for abalone9-18, 0.555 for winequality-red-4, and 0.781 for poker-8 9_vs_6. The decision tree achieved the highest accuracy of 0.984 on the page-blocks0 dataset and the lowest accuracy of 0.555 on the winequality-red-4 dataset. Overall, the classification results demonstrate variability across different datasets and models, underscoring the models' varying applicability and effectiveness. The SVM generally exhibited high accuracy on most datasets, while the decision tree displayed the most fluctuating performance. Logistic regression performed notably well on the ecoli2 dataset.

Figure 6 demonstrates the classification performance of various models under the WGAN-GP oversampling mode. The classification accuracies for logistic regression are as follows: 0.831 for yeast1, 0.957 for ecoli2, 0.863 for page-blocks0, 0.829 for abalone9-18, 0.857 for winequality-red-4, and 0.867 for poker-8 9_vs_6. Logistic regression achieved the highest accuracy of 0.957 on the ecoli2 dataset and the lowest accuracy of 0.829 on the abalone9-18 dataset. For SVMs, the classification accuracies are 0.807 for yeast1, 0.944 for ecoli2, 0.955 for page-blocks0, 0.731 for abalone9-18, 0.739 for winequality-red-4, and 1.000 for poker-8 9_vs_6. SVM achieved perfect accuracy of 1.000 on the poker-8 9_vs_6 dataset, with the lowest accuracy of 0.739 on the winequality-red-4 dataset. The decision tree model yielded the following accuracies: 0.783 for yeast1, 0.979 for ecoli2, 0.994 for page-blocks0, 0.781 for abalone9-18, 0.660 for winequality-red-4, and 1.000 for poker-8 9_vs_6. The decision tree exhibited its highest accuracy of 0.994 on the page-blocks0 dataset and its lowest accuracy of 0.660 on the winequality-red-4 dataset. Overall, the performance of different models under WGAN-GP oversampling varies significantly. Both the SVM and decision tree models achieve perfect accuracy of 1.000 on the poker-8 9_vs_6 dataset, while logistic regression performs best on the ecoli2 dataset with an accuracy of 0.957. The decision tree achieves the highest accuracy of 0.994 on the page-blocks0 dataset but performs poorly on the winequality-red-4 dataset. These results highlight the varying capabilities of different models in handling imbalanced datasets and underscore the impact of WGAN-GP oversampling on model performance.

Table 3 indicates that the WGAN-GP oversampling method generally provides higher classification accuracy compared to the ROS oversampling method across most datasets. Notably, both the SVM and decision tree models achieve perfect accuracy under WGAN-GP oversampling on the poker-8 9_vs_6 dataset. The comparative analysis reveals that WGAN-GP oversampling enhances the performance of classification models on the KEEL dataset, surpassing ROS in classification accuracy. Specifically, logistic regression achieves an average classification accuracy of 86.7%.

B. EFFECTS OF AN INDIVIDUAL RISK PREDICTION MODEL BY THE ENSEMBLE LEARNING AND OVER-SAMPLING

1) MODEL TRAINING OF WGAN-GP OVER-SAMPLING DATASET

The results of the model based on the WGAN-GP oversampling dataset are summarized in Table 4.

Table 4 presents the performance metrics of various models on the WGAN-GP oversampled dataset. In terms of AUC (average), LightGBM achieves the highest score of 0.86, followed by XGBoost (0.79) and GraphSAGE (0.78). LR and Stacking also exhibit respectable AUC values of 0.76 and 0.67 respectively, while DT and GBDT show slightly lower scores of 0.69 and 0.59. For F1-score,

TABLE 4. Results of the model based on WGAN-GP over-sampling.

Model name	AUC (average)	F1-score	Recall	KS (average)
LightGBM	0.86	0.63	0.48	0.87
Extreme Gradient Boosting (XGBoost)	0.79	0.625	0.441	0.438
DT	0.69	0.551	0.45	0.289
LR	0.76	0.62	0.47	0.41
GBDT	0.59	0.53	0.44	0.31
Stacking	0.67	0.59	0.45	0.45
GraphSAGE	0.78	0.61	0.46	0.426

LightGBM and XGBoost perform well with scores of 0.63 and 0.625 respectively, followed by Stacking (0.59) and LR (0.62). GraphSAGE achieves an F1-score of 0.61, while DT and GBDT have lower scores of 0.551 and 0.53. In terms of Recall, LightGBM achieves 0.48, followed by XGBoost (0.441), GraphSAGE (0.46), and LR (0.47). Stacking, DT, and GBDT exhibit lower Recall values of 0.45, 0.45, and 0.44 respectively. Regarding KS (average), LightGBM attains the highest value of 0.87, followed by XGBoost (0.438) and GraphSAGE (0.426). Stacking achieves a KS value of 0.45, while DT and GBDT have lower values of 0.289 and 0.31. Overall, LightGBM outperforms other models across these evaluation metrics, followed by XGBoost and GraphSAGE. DT and GBDT show relatively weaker performance, while Stacking and LR fall within the moderate range. These results suggest that LightGBM is the most suitable classifier choice for the WGAN-GP oversampled dataset, demonstrating its effectiveness in credit risk prediction by balancing precision and recall.

2) MODEL TRAINING WITH THE ORIGINAL DATASET

To assess the efficacy of the WGAN-GP oversampling method, the model is trained using the original dataset. The results of model training on the unaltered dataset are presented in Figure 7.

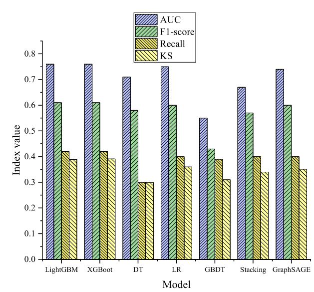

FIGURE 7. Model results based on original data.

Figure 7 illustrates that in comparison to the outcomes presented in Table 4, both LightGBM and XGBoost consistently maintain stable performance in terms of AUC, exhibiting significant enhancements across all metrics with similar magnitudes of improvement. Decision trees notably improve in recall, while logistic regression demonstrates a substantial increase in AUC value, albeit with marginal improvement in recall. The GBDT algorithm, Stacking model, and GraphSAGE model exhibit modest improvements across all metrics, suggesting potential challenges or dataset variations when confronted with new data.

The WGAN-GP oversampling method employed in this study enhances the performance of LightGBM, XGBoost, DT, LR, GBDT algorithm, Stacking model, and GraphSAGE. Robustness, defined as a model's stability against alterations in data or noise, necessitates consideration of its performance under diverse datasets, data distributions, and noise levels. From the available data, an initial evaluation of the robustness of various models can be conducted. LightGBM and XGBoost showcase consistent performance across different datasets, indicating robustness and the ability to maintain effectiveness under varying data distributions. Decision trees exhibit notable improvements in recall but comparatively lower performance in other metrics, hinting at sensitivity to specific data features and potential lower robustness. Logistic regression demonstrates a substantial enhancement in AUC value but limited improvement in recall, suggesting robustness in certain aspects but potential vulnerability in others. The GBDT algorithm, Stacking model, and GraphSAGE model show marginal improvements on new datasets, indicating potential sensitivity to specific data features and suggesting relatively lower robustness.

3) FEATURE IMPORTANCE

Table 5 displays the top 10 important features of the optimal LightGBM model, providing insights into the variables contributing significantly to the likelihood of default. The feature importance ranking results obtained from the experiment are outlined as follows:

TABLE 5. Top 10 important features of the LightGBM model.

Variable name	Feature importance
Annuity_credit_per	0.027
EXT_SOURCE_3	0.02
Days_birth	0.017
EXT_SOURCE_2	0.016
EXT_SOURCE_1	0.0155
AMT_ annuity	0.013
DAYS_ID_PUBLISH	0.012
AMT_ credit	0.007
Annuity_income_per	0.006
Work_birth_per	0.0056

In Table 5, 'Annuity_credit_per' signifies the proportion of the user's monthly payment to the total credit loan. 'Days_birth' represents the number of days between the applicant's date of birth and a reference date. 'EXT_SOURCE_1', 'EXT_SOURCE_2', and 'EXT_SOURCE_3' are external data sources containing information about the candidate's credit history, utilized in assessing the applicant's credit risk. 'AMT_annuity' and 'DAYS_ID_PUBLISH' indicate the duration between the date the applicant publishes the ID file and a reference date. 'AMT_credit' refers to the total amount of credit, while 'Annuity_income_per' signifies the proportion of the annuity to the applicant's income. Lastly, 'Work_birth_per' indicates the loan years to age ratio. The data in Table 5 reveal that several key features exert a significant impact on the model's prediction of the likelihood of default. Among these, 'Annuity_credit_per' (annuity to credit limit ratio) holds substantial feature importance, indicating its pivotal role in assessing the likelihood of default. Additionally, 'AMT_annuity' and 'AMT_credit' (annuity and line of credit) also contribute to predicting the likelihood of default.

4) MODEL RESULTS BASED ON REAL PERSONAL CREDIT DATA

To further validate the proposed model, real personal credit data from 498 domestic commercial banks, sourced from the Tianchi database, was utilized. The data has been desensitized to ensure privacy protection while maintaining authenticity. Figure 8 presents the results of the model applied to this real personal credit data.

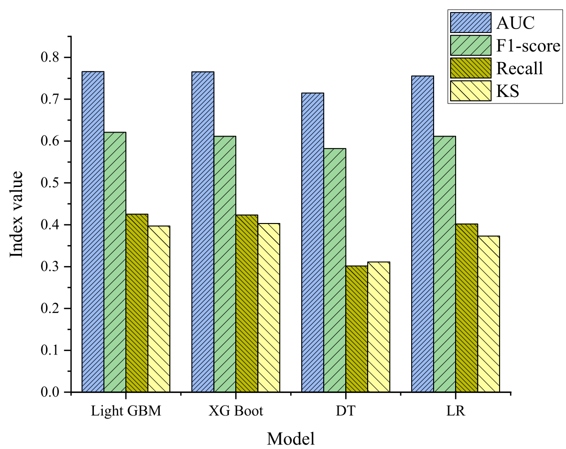

FIGURE 8. Model results based on real personal credit data.

Figure 8 illustrates the performance of four models evaluated on real personal credit data from 498 domestic commercial banks. The LightGBM model achieved the highest overall performance, with an AUC of 0.766, an F1-score of 0.621, a Recall of 0.425, and a KS value of 0.397, demonstrating a strong balance between accuracy and discrimination.

The XGBoost model, with an identical AUC of 0.766, exhibited a slightly lower F1-score of 0.611, but achieved a Recall of 0.423 and a KS value of 0.403, indicating robust discriminatory capabilities. The decision tree model showed an AUC of 0.715, an F1-score of 0.582, a Recall of 0.302, and a KS value of 0.311, reflecting relatively weaker performance but still offering practical utility in certain contexts. The logistic regression model presented balanced classification abilities, with an AUC of 0.756, an F1-score of 0.611, a Recall of 0.402, and a KS value of 0.373. Overall, the LightGBM model demonstrated superior prediction accuracy and risk discrimination compared to the other models, and these findings are consistent with results obtained from the original data analysis.

C. DISCUSSION

Through a series of in-depth experiments and analyses, this study provides a comprehensive discussion on the effectiveness of the WGAN-GP model for data oversampling. Firstly, the observed loss situation of the WGAN-GP model, as indicated by experimental results, reveals that the discriminator loss tends to approach 0, signifying the successful discrimination between the distribution of generated samples and real samples by the discriminator. Regarding classification performance, a comparison of classification results among different models under various oversampling methods highlights the superior performance of the WGAN-GP oversampling method, particularly on the KEEL dataset. Compared to the ROS oversampling method, WGAN-GP significantly enhances the classification models' performance, achieving a remarkable average classification accuracy of 86.7% under the LR method. Furthermore, model training results based on the WGAN-GP oversampling dataset demonstrate outstanding performance of the LightGBM model across multiple metrics such as AUC, F1-score, Recall, and KS value, showcasing exceptional predictive capabilities. The effectiveness of the WGAN-GP oversampling method is validated through a comparison with the dataset without oversampling. Finally, through the analysis of feature importance, several key features crucial for predicting the likelihood of default are identified. Among these, "Annuity_credit_per," "EXT_SOURCE_3," and "Days_birth" offer information closely related to credit risk, emphasizing their significant roles in the model's predictive accuracy.

V. CONCLUSION

In response to the escalating demand for personal credit assessment, the establishment of an accurate and efficient credit risk assessment system has become a pressing concern for banking and financial institutions. This study addresses challenges related to credit data imbalance and strives to construct robust predictive models, making noteworthy contributions in various aspects. Firstly, the introduction of the WGAN-GP oversampling method aims to rectify data imbalances by synthesizing default samples. In contrast to conventional oversampling methods, WGAN-GP more precisely simulates real data distributions, effectively mitigating data imbalance issues. Application of this method not only enhances model performance but also bolsters its robustness. Secondly, this study integrates GAN with the ensemble learning model LightGBM to formulate a credit risk prediction model. By leveraging oversampling data generated by GAN and harnessing the advantages of LightGBM, a more accurate and reliable predictive model is established. This approach, fusing different methods, introduces novel ideas and approaches to credit risk prediction. Finally, experimental results underscore the exemplary performance of the proposed credit risk prediction model based on the LightGBM-GAN framework across multiple evaluation metrics, including AUC, F1-score, Recall, and KS value. This accomplishment furnishes banking and financial institutions with a dependable and effective credit risk assessment tool, aiding in risk reduction and enhancing operational efficiency and risk management standards. In summary, this study tackles the challenges of credit data imbalance and constructing robust predictive models and furnishes dependable credit risk assessment solutions for financial institutions. This holds significance in reducing financial institution risk costs, improving the inclusiveness and sustainability of financial services, and making positive contributions to the development of the financial industry and economic stability.

One limitation of this study pertains to the use of a limited dataset. To address this, future research will focus on expanding the scope of experiments and validating with more diverse datasets to ensure the model's universality and robustness. Additionally, further research can delve into enhancing model interpretability to better understand predictive mechanisms and provide more convincing explanations for applications in different industries. Moreover, by actively involving more domain experts in research and leveraging their expertise to improve and optimize the model to meet the practical needs of financial institutions, reliable credit risk assessment solutions can be provided for a wider range of financial domains, making greater contributions to the stability and risk control of the financial system.

DATA AVAILABILITY STATEMENTS

All data generated or analysed during this study are included in this published article [and its supplementary information files].

CONFLICT OF INTEREST STATEMENT

This study does not have competing interests as defined by nature research, or other interests that may be considered to influence the results reported and discussed herein.

REFERENCES

[1] M. Brogi, V. Lagasio, and P. Porretta, "Be good to be wise: Environmental, social, and governance awareness as a potential credit risk mitigation factor," *J. Int. Financial Manage. Accounting*, vol. 33, no. 3, pp. 522–547, Oct. 2022.

[2] J. P. Hughes, J. Jagtiani, and C.-G. Moon, "Consumer lending efficiency: Commercial banks versus a fintech lender," *Financial Innov.*, vol. 8, no. 1, pp. 1–39, Dec. 2022.

[3] M. Naili and Y. Lahrichi, "The determinants of banks' credit risk: Review of the literature and future research agenda," *Int. J. Finance Econ.*, vol. 27, no. 1, pp. 334–360, Feb. 2022.
[4] X. Y. Zhou, B. Caldecott, A. G. F. Hoepner, and Y. Wang, "Bank green lending and credit risk: An empirical analysis of China's green credit policy," *Bus. Strategy Environ.*, vol. 31, no. 4, pp. 1623–1640, May 2022.
[5] G. R. Murodovich and I. A. Komolboevich, "Problems of population banking of commercial banks," *Int. J. Integr. Educ.*, vol. 5, no. 3, pp. 91–95, Feb. 2022.
[6] M. Foglia, "Non-performing loans and macroeconomics factors: The Italian case," *Risks*, vol. 10, no. 1, p. 21, Jan. 2022.
[7] I. A. Zahid and S. Sabbar Joudar, "Does lack of knowledge and hardship of information access signify powerful AI? A large language model perspective," *Appl. Data Sci. Anal.*, vol. 2023, pp. 150–154, Dec. 2023.
[8] S. P. Porkodi, V. Sarada, V. Maik, and K. Gurushankar, "Generic image application using GANs (Generative adversarial Networks): A review," *Evolving Syst.*, vol. 14, no. 5, pp. 903–917, Oct. 2023.
[9] Z. Li, M. Usman, R. Tao, P. Xia, C. Wang, H. Chen, and B. Li, "A systematic survey of regularization and normalization in GANs," *ACM Comput. Surv.*, vol. 55, no. 11, pp. 1–37, Nov. 2023.
[10] A. S. Albahri, I. A. Zahid, M. G. Yaseen, M. Aljanabi, A. H. Ali, and A. Kaleel, "Harnessing the tide of innovation: The dual faces of generative AI in applied sciences; letter to editor," *Appl. Data Sci. Anal.*, vol. 2024, pp. 1–3, Jan. 2024.
[11] M. Li and Y. Fu, "Prediction of supply chain financial credit risk based on PCA-GA-SVM model," *Sustainability*, vol. 14, no. 24, p. 16376, Dec. 2022.
[12] S. Shi, R. Tse, W. Luo, S. D'Addona, and G. Pau, "Machine learning-driven credit risk: A systemic review," *Neural Comput. Appl.*, vol. 34, no. 17, pp. 14327–14339, Sep. 2022.
[13] M. F. Shams, H. Jahangirnia, and R. Gh. Jamkarani, "Designing credit risk early-warning system for individual and corporate customers of the bank using multiple logit comparison model and survival function," *Int. J. Finance Managerial Accounting*, vol. 7, no. 25, pp. 163–177, Feb. 2022.
[14] G. Yao, X. Hu, T. Zhou, and Y. Zhang, "Enterprise credit risk prediction using supply chain information: A decision tree ensemble model based on the differential sampling rate, synthetic minority oversampling technique and AdaBoost," *Expert Syst.*, vol. 39, no. 6, p. 12953, Feb. 2022.
[15] P. Pandey and K. C. Bandhu, "A credit risk assessment on borrowers classification using optimized decision tree and KNN with Bayesian optimization," *Int. J. Inf. Technol.*, vol. 14, no. 7, pp. 3679–3689, Dec. 2022.
[16] A. Gnoatto, A. Picarelli, and C. Reisinger, "Deep xVA solver: A neural network–based counterparty credit risk management framework," *SIAM J. Financial Math.*, vol. 14, no. 1, pp. 314–352, Oct. 2022.
[17] S.-Y. Lin, D.-R. Liu, and H.-P. Huang, "Credit default swap prediction based on generative adversarial networks," *Data Technol. Appl.*, vol. 56, no. 5, pp. 720–740, Dec. 2022.
[18] N. Park, Y. H. Gu, and S. J. Yoo, "Synthesizing individual consumers' credit historical data using generative adversarial networks," *Appl. Sci.*, vol. 11, no. 3, p. 1126, Jan. 2021.
[19] S. Mahajan, A. Nayyar, and A. Raina, "A Gaussian process-based approach toward credit risk modeling using stationary activations," *Concurrency Comput., Pract. Exper.*, vol. 34, no. 5, p. 6692, Feb. 2022.
[20] L. He, S. Gan, and T. Zhong, "The impact of green credit policy on firms' green strategy choices: Green innovation or green-washing?" *Environ. Sci. Pollut. Res.*, vol. 29, no. 48, pp. 73307–73325, Feb. 2022.
[21] X. Fan, X. Guo, Q. Chen, Y. Chen, T. Wang, and Y. Zhang, "Data augmentation of credit default swap transactions based on a sequence GAN," *Inf. Process. Manage.*, vol. 59, no. 3, May 2022, Art. no. 102889.
[22] S. Lin and A. C. Wang, "Self-attention recurrent conditional generative adversarial networks for corporate credit rating prediction," *J. Inf. Sci. Eng.*, vol. 39, no. 5, pp. 1209–1230, Oct. 2022.
[23] L. Huang, L. Li, X. Wei, and D. Zhang, "Short-term prediction of wind power based on BiLSTM–CNN–WGAN-GP," *Soft Comput.*, vol. 26, no. 20, pp. 10607–10621, Oct. 2022.
[24] C. Sun, X. Zhang, H. Meng, X. Cao, and J. Zhang, "AC-WGAN-GP: Generating labeled samples for improving hyperspectral image classification with small-samples," *Remote Sens.*, vol. 14, no. 19, p. 4910, Oct. 2022.
[25] L. Peng, S. Li, and H. Sun, "A pipe ultrasonic guided wave signal generation network suitable for data enhancement in deep learning: U.S.-WGAN," *Energies*, vol. 15, no. 18, p. 6695, Feb. 2022.
[26] W. Hou, H. Guo, B. Yan, Z. Xu, C. Yuan, and Y. Mao, "Tool wear state recognition under imbalanced data based on WGAN-GP and lightweight neural network ShuffleNet," *J. Mech. Sci. Technol.*, vol. 36, no. 10, pp. 4993–5009, Oct. 2022.
[27] Q. Li, D. Zhang, and K. Yan, "A solar irradiance forecasting framework based on the CEE-WGAN-LSTM model," *Sensors*, vol. 23, no. 5, p. 2799, Mar. 2023.
[28] C. Jia, P. Wei, Z. Lu, M. Ye, R. Zhu, and H. Liu, "A novel prediction approach of polymer gear contact fatigue based on a WGAN-XGBoost model," *Fatigue Fract. Eng. Mater. Struct.*, vol. 46, no. 6, pp. 2272–2283, Jun. 2023.
[29] Y. Lu, X. Tao, N. Zeng, J. Du, and R. Shang, "Enhanced CNN classification capability for small Rice disease datasets using progressive WGAN-GP: Algorithms and applications," *Remote Sens.*, vol. 15, no. 7, p. 1789, Mar. 2023.
[30] L. Yang, H. Sun, J. Zhang, S. Mo, W. Jiang, and X. Zhang, "DFS-GAN: Stabilizing training of generative adversarial networks through discarding fake samples," *J. Electron. Imag.*, vol. 31, no. 6, Nov. 2022, Art. no. 063016.
[31] F. Coelho, M. F. Pinto, A. G. Melo, G. S. Ramos, and A. L. M. Marcato, "A novel sEMG data augmentation based on WGAN-GP," *Comput. Methods Biomech. Biomed. Eng.*, vol. 26, no. 9, pp. 1008–1017, Jul. 2023.
[32] S. Yean, W. Goh, B.-S. Lee, and H. L. Oh, "ExtendGAN+: Transferable data augmentation framework using WGAN-GP for data-driven indoor localisation model," *Sensors*, vol. 23, no. 9, p. 4402, Apr. 2023.
[33] J.-T. Li, Z. Bian, and L.-X. Guo, "Optimized complex object classification model: Reconstructing the ISAR image of a hypersonic vehicle covered with a plasma sheath using a U-WGAN-GP framework," *Int. J. Remote Sens.*, vol. 43, no. 14, pp. 5306–5323, Jul. 2022.
[34] E. Hashimoto-Roth, A. Surendra, M. Lavallée-Adam, S. A. L. Bennett, and M. Čuperlović-Culf, "METAbolomics data balancing with over-sampling algorithms (META-BOA): An online resource for addressing class imbalance," *Bioinformatics*, vol. 38, no. 23, pp. 5326–5327, Nov. 2022.
[35] S. Kumar, A. Mallik, and B. S. Panda, "Link prediction in complex networks using node centrality and light gradient boosting machine," *World Wide Web*, vol. 25, no. 6, pp. 2487–2513, Feb. 2022.

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